# BOUNDED HOMOTOPY CATEGORIES ARE TRIANGULATED SUBCATEGORIES

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*Editorial note.* Counted span is the author's own proof environment, source lines 3625-3645, following the statement at 3618-3623.

*Editorial note.* No omissions: the cone boundedness is read off from the construction; a second complete argument is appended.

*Allowed standard prerequisites (editorial).*

- Triangulated subcategory criterion. lemma-triangulated-subcategory (same chapter, cited at ~3631). Reduces to shift preservation and cones staying inside.
- Alternative route. lemma-make-injective (carded L26206, cited at ~3643). The author offers a second argument via termwise split injections.

*Source reference.* Original lemma 013N: sources/stacks-derived/pinned-original-source.tex, source lines 2840–2840. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

*Source reference.* Original lemma 05QX: sources/stacks-derived/pinned-original-source.tex, source lines 1077–1077. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

**Lemma 0.1.** *Let $\mathcal{A}$ be an additive category. The categories $K^+(\mathcal{A})$, $K^-(\mathcal{A})$, and $K^b(\mathcal{A})$ are full triangulated subcategories of $K(\mathcal{A})$.*

*Proof.* Each of the categories mentioned is a full additive subcategory. We use the criterion of Lemma 05QX to show that they are triangulated subcategories. It is clear that each of the categories $K^+(\mathcal{A})$, $K^-(\mathcal{A})$, and $K^b(\mathcal{A})$ is preserved under the shift functors $[1], [-1]$. Finally, suppose that $f : A^\bullet \to B^\bullet$ is a morphism in $K^+(\mathcal{A})$, $K^-(\mathcal{A})$, or $K^b(\mathcal{A})$. Then $(A^\bullet, B^\bullet, C(f)^\bullet, f, i, -p)$ is a distinguished triangle of $K(\mathcal{A})$ with $C(f)^\bullet \in K^+(\mathcal{A})$, $K^-(\mathcal{A})$, or $K^b(\mathcal{A})$ as is clear from the construction of the cone. Thus the lemma is proved. (Alternatively, $K^\bullet \to L^\bullet$ is isomorphic to an termwise split injection of complexes in $K^+(\mathcal{A})$, $K^-(\mathcal{A})$, or $K^b(\mathcal{A})$, see Lemma 013N and then one can directly take the associated distinguished triangle.) $\square$

*Date*: stacks-project:derived.tex:local-snapshot-20261012; GNU FDL 1.2 or later.